

Ashwin K. Aggarwal

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EDUCATION

Cornell University, College of Engineering

Ithaca, NY

Bachelor of Science in Computer Science, Artificial Intelligence Minor

Relevant Coursework: Linear Algebra, Data Structures (Java), Functional Programming (OCaml), Probability and Stats II, Discrete Math, Algorithms, Computer Systems, Large-Scale Machine Learning, Computer Vision, Robotics, Deep Learning

EXPERIENCE

Virginia Commonwealth University

May 2026 – Present

Medical Machine Learning Research Assistant, Wargo Lab

Richmond, VA

- Implemented ETL pipelines for imaging machines, automating PowerBI dashboard generation for medical imaging; scheduled workflow for concurrently filling a SQL Server database to replace manual data collection at 5+ hospitals
- Engineered PyTorch Computer Vision model turning 2D mammography images to 3D tomosynthesis simulations on 500+ breast scans using CUDA parallelization; used Monte Carlo simulation to determine lesion position probabilities
- Co-authored a manuscript documenting the system design, automation workflow, and architecture for publication

Virginia Commonwealth University

May 2026 – Jun 2026

Data Engineering Intern, AI Department

Remote

- Designed Databricks pipeline for processing large datasets of unstructured, sensitive clinical text files from an Epic database, sorting by medallion architecture to optimize RAG performance on downstream medical queries
- Leveraged OCR models to extract 100+ fields in a clinical note documents to ingest into agent for categorizing demographics and diagnosis; cut LLM API usage by 90% through prompt engineering and structured retrieval

Cornell Computational Biology Lab

Jan 2026 – May 2026

Undergraduate Machine Learning Researcher

Ithaca, NY

- Constructed and fine-tuned 3D CNNs (ResNet and U-Net architectures) for classifying parity of 500+ MRI scans; improved F1 Score (0.54 to 0.61) and overall accuracy using data augmentation, dropout, early stopping in PyTorch
- Created data pipelines normalize structural brain scans, iterating over 10+ model architectures to probe signal

Next Play

Jul 2025 – Aug 2025

Software Engineering Intern, AI/ML Team

Remote

- Architected a real-time Computer Vision pipeline in TypeScript and React using MediaPipe pose detection, supporting rep counting and live feedback across 20+ fitness challenges, achieving 90% pose classification accuracy for users
- Developed and shipped 10+ backend features and REST API integrations for a sports game in Golang and Kotlin, deployed on AWS, contributing to gameplay functionality and cross-service communication at a fast-moving startup

PROJECTS

AI Housing Lease Analyzer | *Python (OpenAI API, LangChain, FAISS, BM25), Docker*

- Built a Hybrid RAG pipeline combining FAISS vector search and BM25 retrieval with LangChain and OpenAI to parse and reason over complex multi-page housing leases, achieving 94% faithfulness via RAGAS evaluation
- Launched a Streamlit app enabling natural language contract queries, cutting manual lease review workload by 80%

Multi-Agent Research Assistant | *Docker, Python (Claude API, Neo4j, APScheduler, LangGraph), pyvis*

- Created a four-agent system on the Claude API that ingests, summarizes, and cross-links research papers into a interactive Neo4j knowledge graph, surfacing 100+ connections across 30+ papers missed by keyword search
- Automated nightly ingestion with APScheduler and shipped a Streamlit interface with pyvis graph visualization, letting users trace citation and concept paths across the corpus in place of manual literature review

AI Basketball Shot Tracker | *Python (Numpy, Matplotlib, OpenCV), Ultralytics (YOLO), RoboFlow*

- Fine-tuned a YOLOv8 model on 300+ hand-labeled frames for ball and hoop detection, achieving 97% accuracy using RoboFlow for dataset annotation, augmentation, and tracking across dozens of model training and evaluation runs
- Classified made and missed shots using live ball and hoop position tracking via OpenCV, achieving 90%+ accuracy

SKILLS/INTERESTS

Skills: Java, Python, C#, C, C++, PyTorch, Docker, CI/CD, Kubernetes, CUDA, TypeScript, Go, Rust, AWS, ROS

Interests: Catan, Physical AI, MarioKart, Philosophy, Basketball, New England Patriots, Golf, Pickleball, Chess, Startups